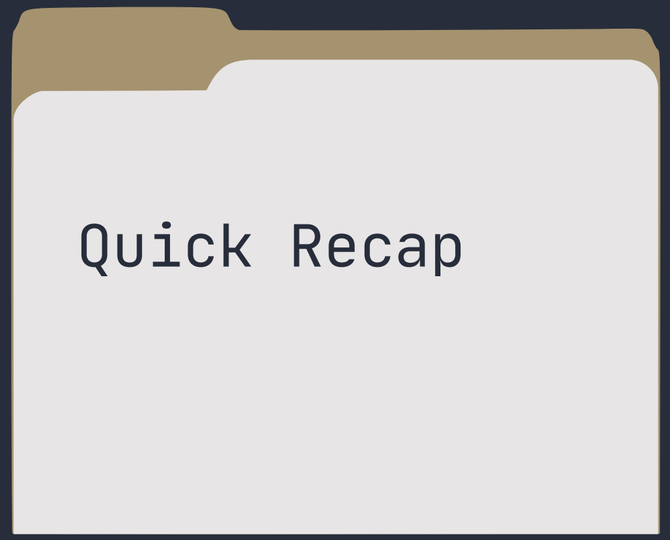
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AI School Iteration 2.4

Classification Metrics

<h4> By BOULGHERAIF Fatma Zohra </h4>

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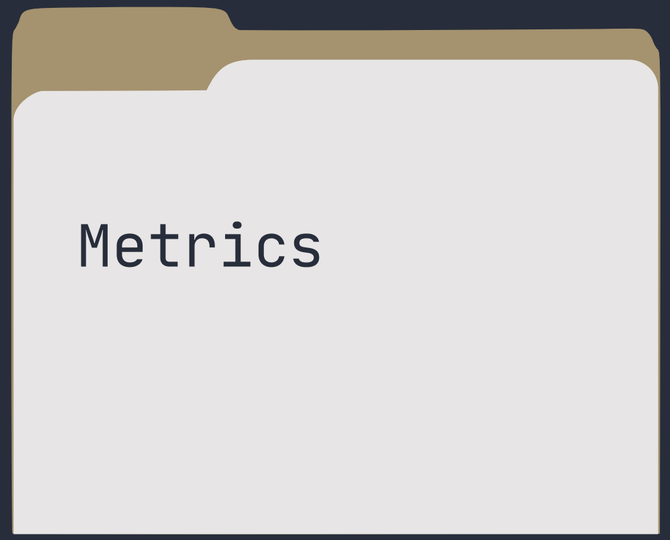
01. What Classification Outputs

Two kinds of output depending on the model:

Output type	Example	Used for
Probability	P(y=1 | x) = 0.73	Logistic regression
Label	ŷ = 1	Decision trees, kNN

When you have probabilities, a threshold converts them to a label:
ŷ = 1 if P(y=1 | x) ≥ threshold
ŷ = 0 otherwise

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02. Accuracy

Proportion of correctly classified samples over the total number of samples.

$$\text{Accuracy} = \#\ \text{Correct predictions} \ / \ \#\ \text{Total predictions}$$

Limitation – class imbalance:

Consider a dataset where 99% of samples belong to class 0. A classifier that always predicts class 0 achieves 99% accuracy while never detecting class 1

Accuracy is an insufficient metric when:
- Class distributions are **imbalanced**
- The **cost** of false positives and false negatives differs

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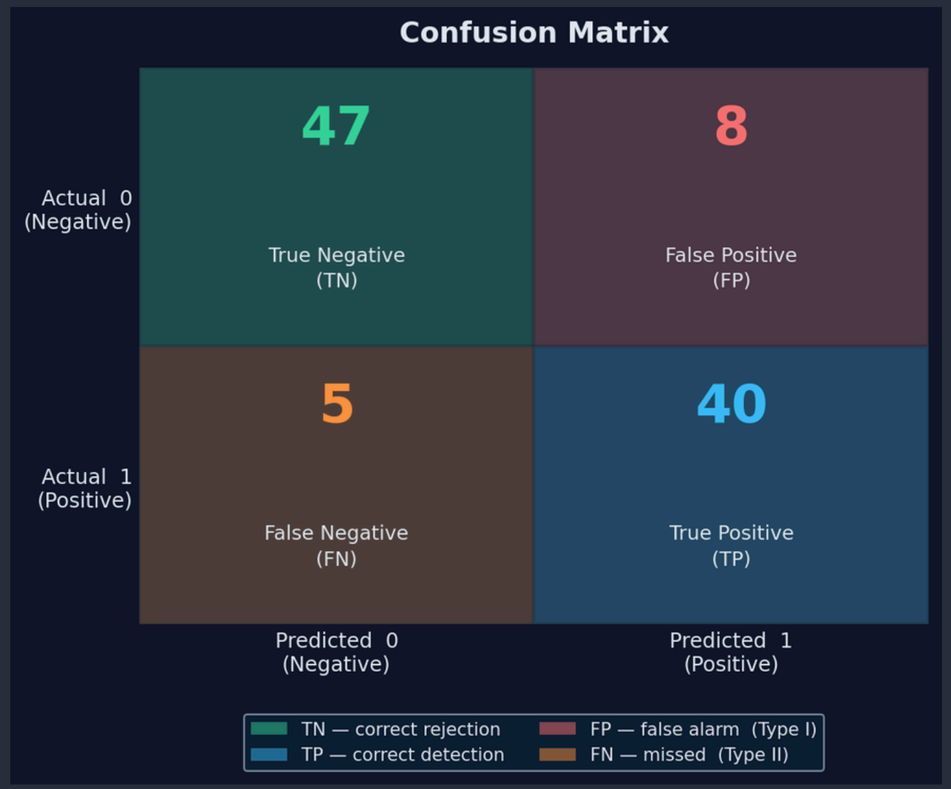
03. Confusion Matrix

For binary classification, every prediction falls into one of four categories: FP, TP, FN, TN

Definitions:
- TP: positive sample correctly classified as positive
- TN: negative sample correctly classified as negative
- FP: negative sample incorrectly classified as positive
- FN: positive sample incorrectly classified as negative

F => incorrectly classified as P or N
T => correctly classified as P or N

03. Confusion Matrix



The relative cost of FP vs. FN is problem-dependent and should be defined before model selection.

Examples:

- Medical diagnosis: FN (missed disease) is typically more costly than FP
- Spam filtering: FP (legitimate email classified as spam) may be more costly than FN

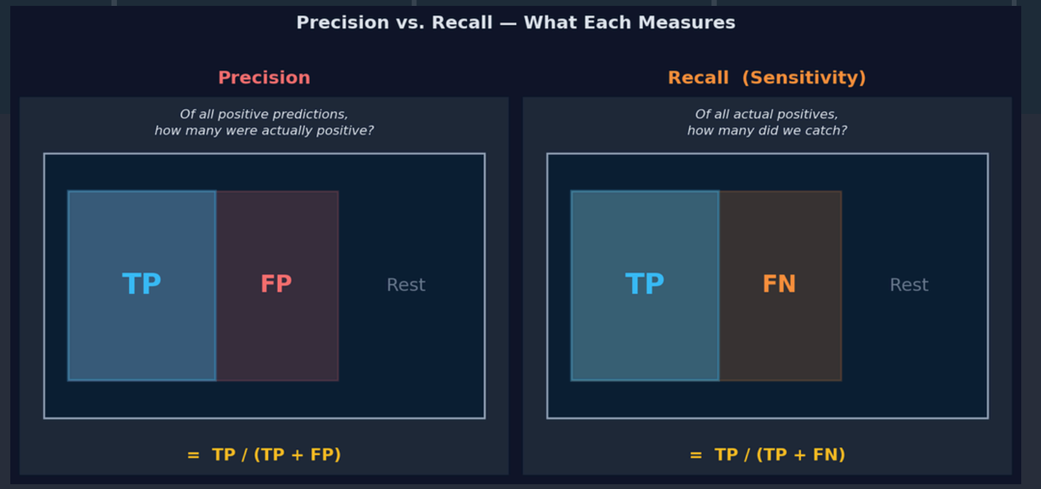
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02. Accuracy

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

04. Precision & Recall



- **Precision:** Proportion of positive predictions that are correct. Quantifies the false alarm rate.

$$\text{Precision (P)} = TP / (TP + FP)$$

AKA TP / Prediction = 1

- **Recall (True Positive Rate / Sensitivity):** Proportion of actual positives that were correctly detected. Quantifies the missed detection rate.

$$\text{Recall (R)} = TP / (TP + FN)$$

AKA TP / Real = 1

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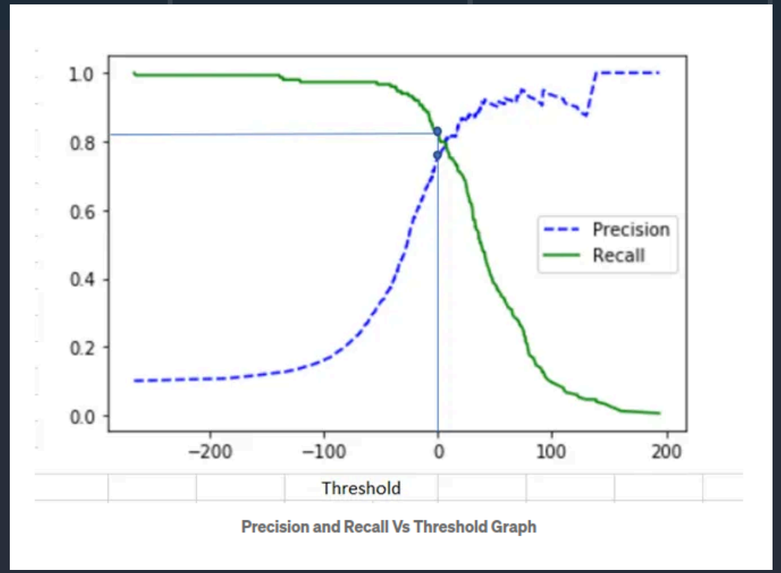
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04. Precision & Recall

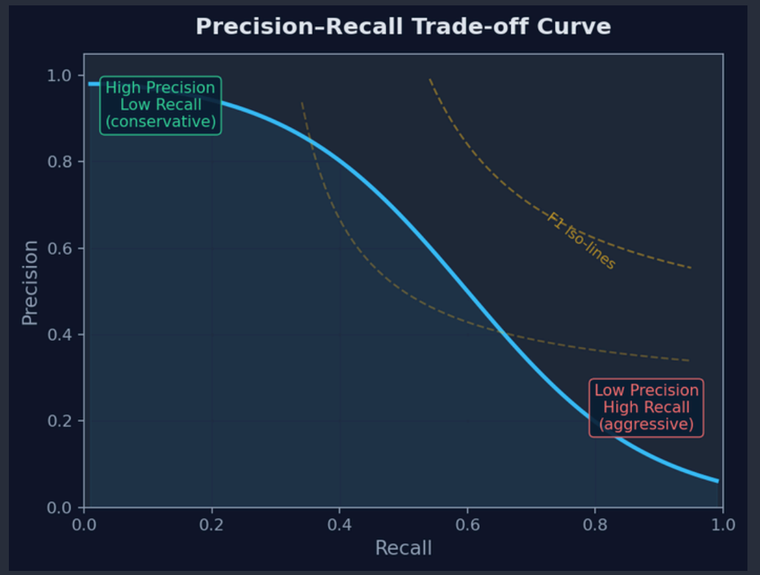
The trade-off in the case of probability based classification:

Pushing the threshold down (more aggressive):
- ↑ Recall – you catch more positives
- ↓ Precision – but you also flag more negatives wrongly

Pushing the threshold up (more conservative):
- ↑ Precision – everything you flag is likely real
- ↓ Recall – but you miss more actual positives



Precision and Recall Vs Threshold Graph



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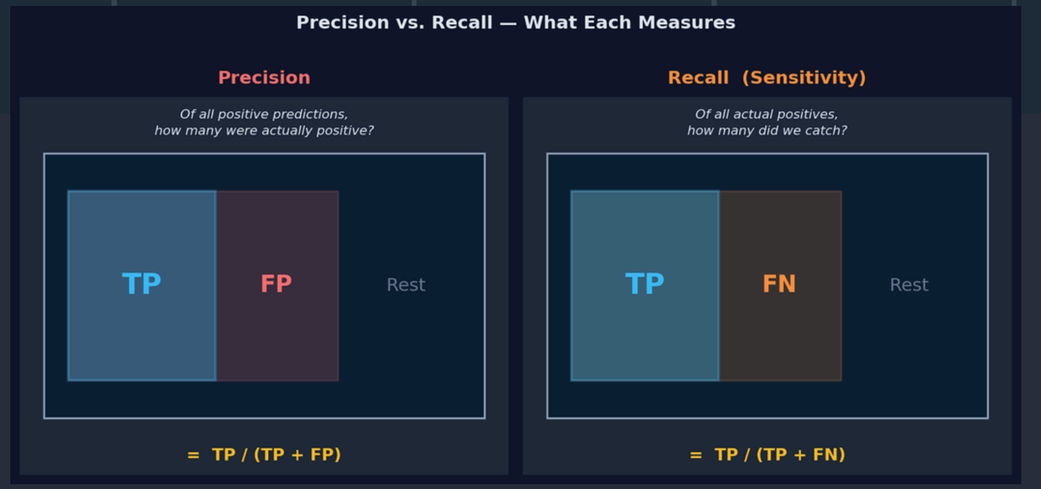
05. F1 Score

F1 is the harmonic mean of precision and recall:

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall)$$

The harmonic mean is used rather than the arithmetic mean because it returns 0 when either component is 0. A degenerate classifier predicting all samples as positive achieves Recall = 1 but Precision ≈ 0, yielding F1 ≈ 0.

05. Fβ Score

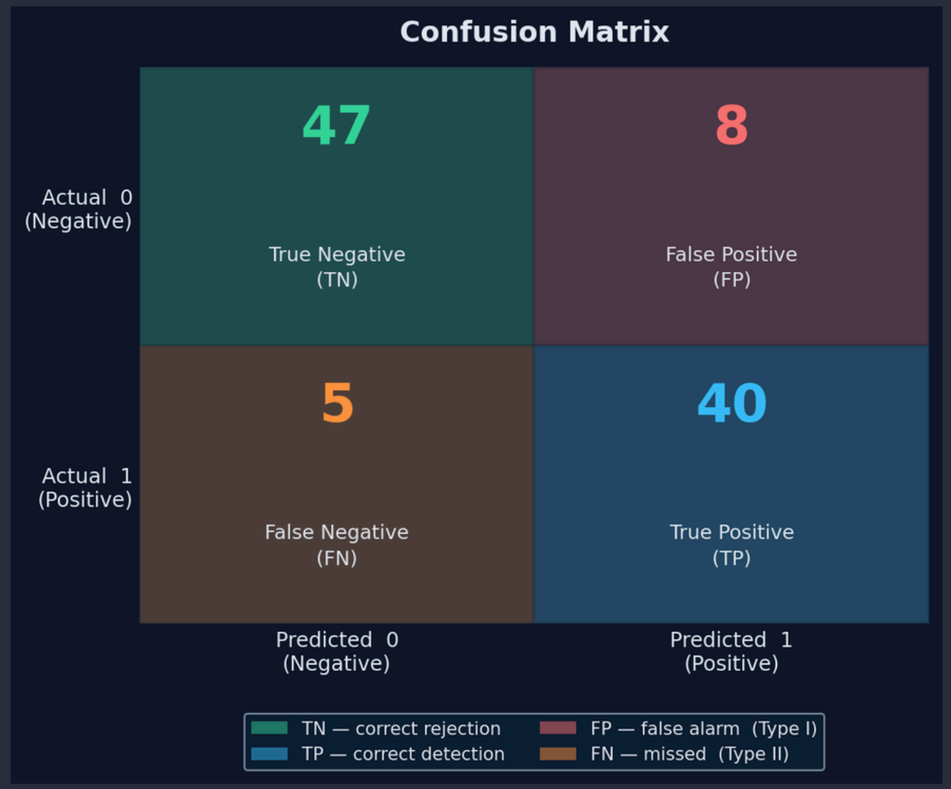


F-beta score: generalisation allowing asymmetric weighting:

$$F\beta = (1 + \beta^2) \cdot (\text{Precision} \cdot \text{Recall}) / (\beta^2 \cdot \text{Precision} + \text{Recall})$$

beta	weighting
β < 1	Precision-weighted: penalises FP more heavily
β = 1	Standard F1: equal weighting
β > 1	Recall-weighted: penalises FN more heavily

Practice Time



- Accuracy?
- Precision?
- Recall?
- F1 Score

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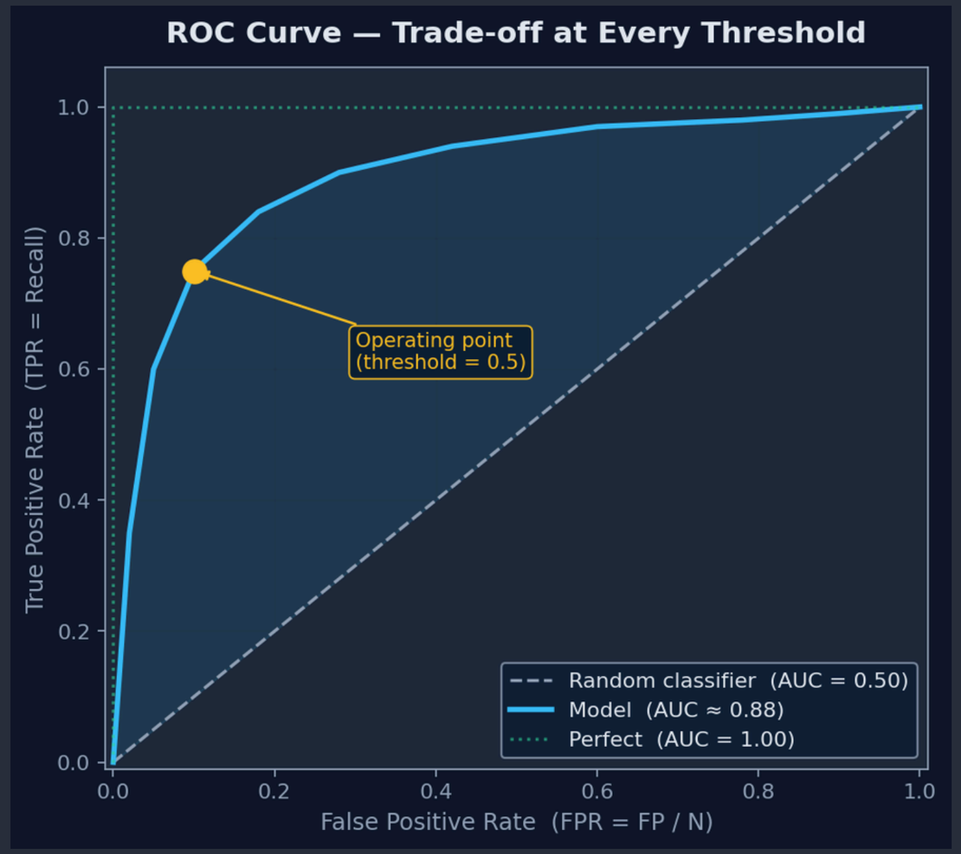


06. ROC Curve & AUC (1/2)

Rather than evaluating at a single threshold, the ROC curve characterises classifier performance across all thresholds $\tau \in [0, 1]$.

+ Threshold-independent: evaluates the model's ability to rank positive samples above negative ones

x-axis: FPR = FP / (FP + TN)
y-axis: TPR = TP / (TP + FN)

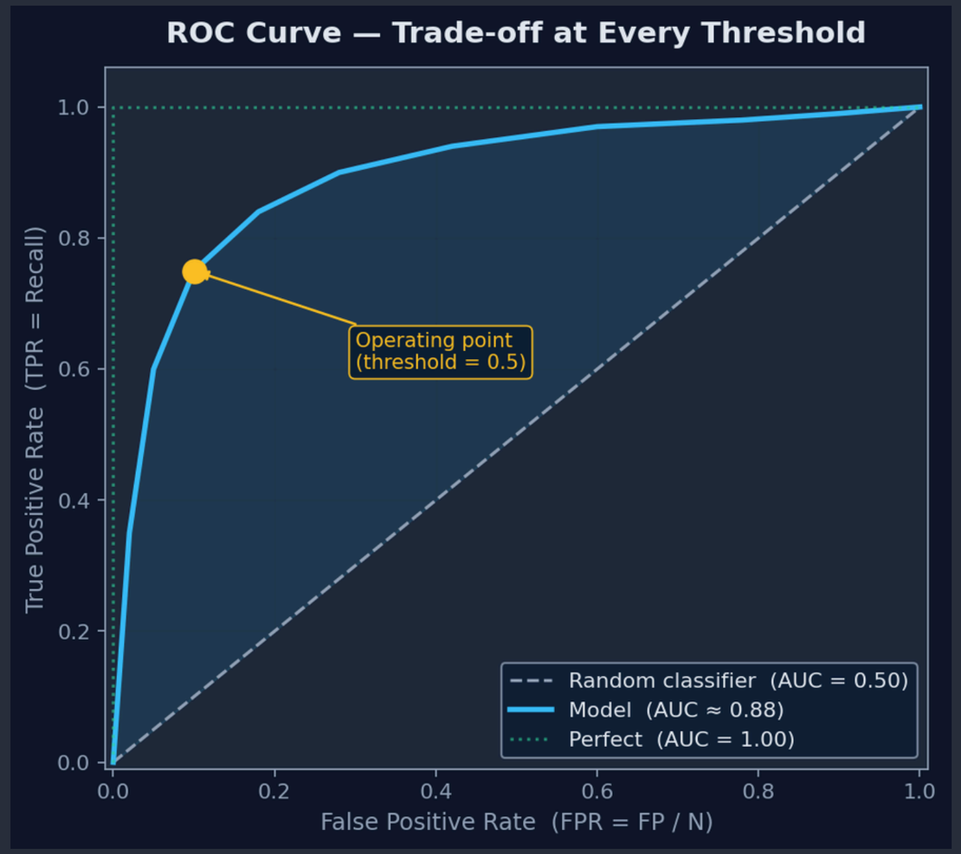


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06. ROC Curve & AUC (2/2)

AUC (Area Under the ROC Curve)

AUC	Interpretation
1.0	Perfect classifier
0.5	**Random** classifier, no discriminative power
<0.5	Worse than random



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Multiclass classification

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10. Multiclass Classification

Definition: Classification where $y \in \{1, 2, ..., K\}$ for $K > 2$.

Examples:
- Iris species → {Setosa, Versicolor, Virginica}
- Image recognition → {Cat, Dog, Bird, Fish, Horse, ...}
- Clinical staging → {Benign, Stage I, Stage II, Stage III}

Main approaches:
- Native multiclass: Model outputs K probabilities simultaneously — **Softmax**
- Some models (like logistic regression or SVMs) handle multiclass by combining multiple binary classifiers (**OvR / OvO**).
- Many modern models support multiclass directly.



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11. Softmax for Multiclass Output (1/3)

For K classes, the model computes **K independent linear scores** (logits), one for each class:

$$z_k = \beta_{0k} + \beta_{1k}x_1 + \ldots + \beta_{nk}x_n \qquad \text{for } k = 1, \ldots, K$$

The **Softmax** function normalises the logit vector into a valid probability distribution:

$$P(y = k \mid x) = \exp(z_k) \,/\, \Sigma_j \exp(z_j)$$

- $P(y = k \mid x) \in (0, 1)$ for all k
- $\Sigma_k P(y = k \mid x) = 1$

11. Softmax for Multiclass Output (2/3)

Prediction:

$$\hat{y} = \text{argmax}_k \, P(y = k \mid x)$$

The sigmoid function is a special case of Softmax with K = 2.

```
import numpy as np

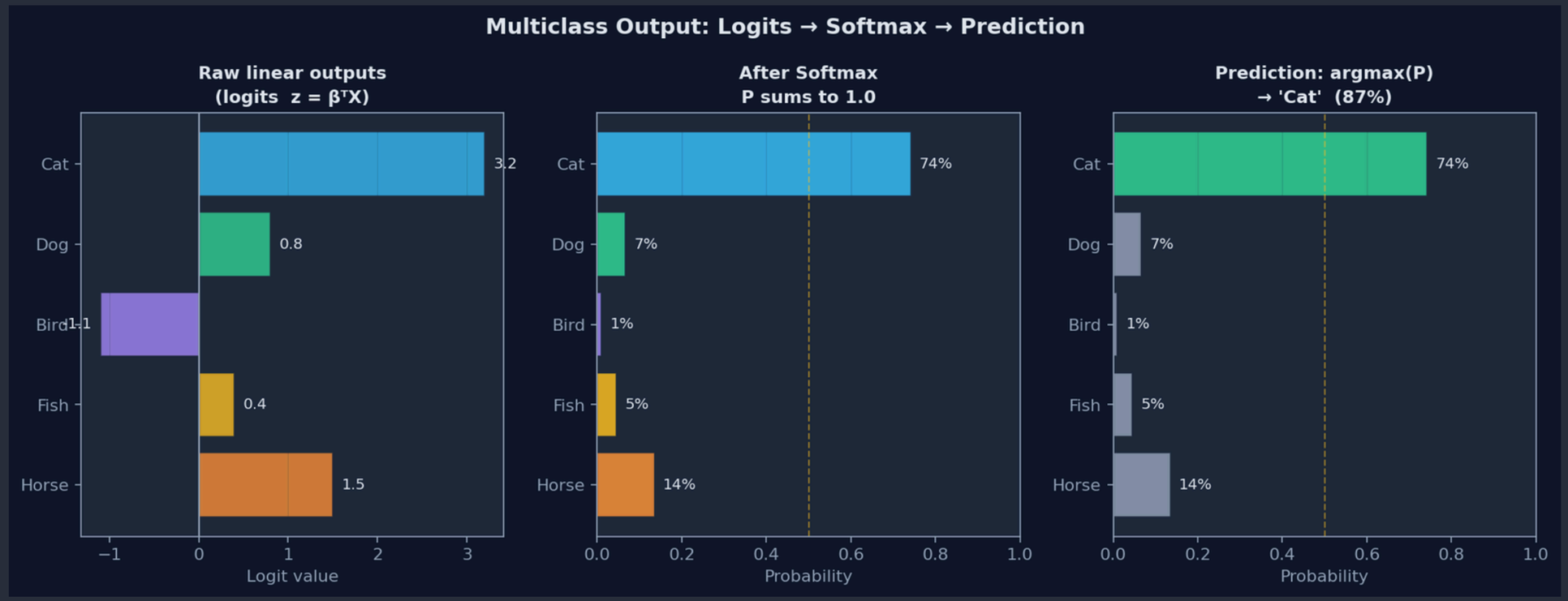
ls = [1, 2, 4, 5, 8, 3]

print(f"The maximum is located at: {np.argmax(ls)}. Its value is: {ls[np.argmax(ls)]}")
```

```
The maximum is located at: 4. Its value is: 8
```

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11. Softmax for Multiclass Output (3/3)

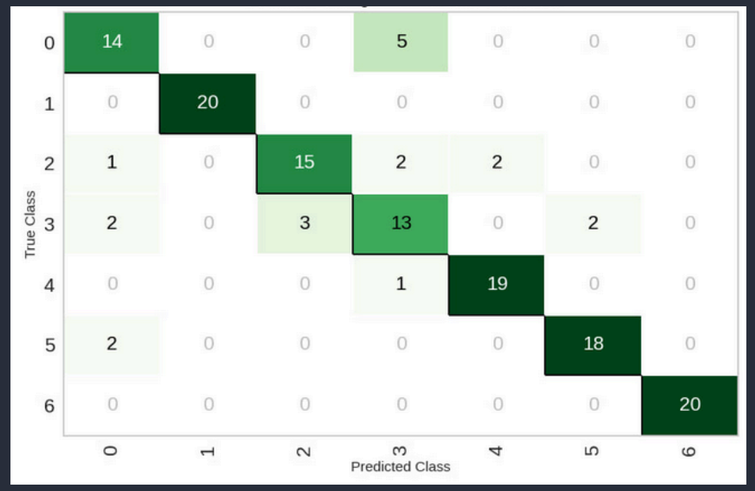


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Multiclass evaluation

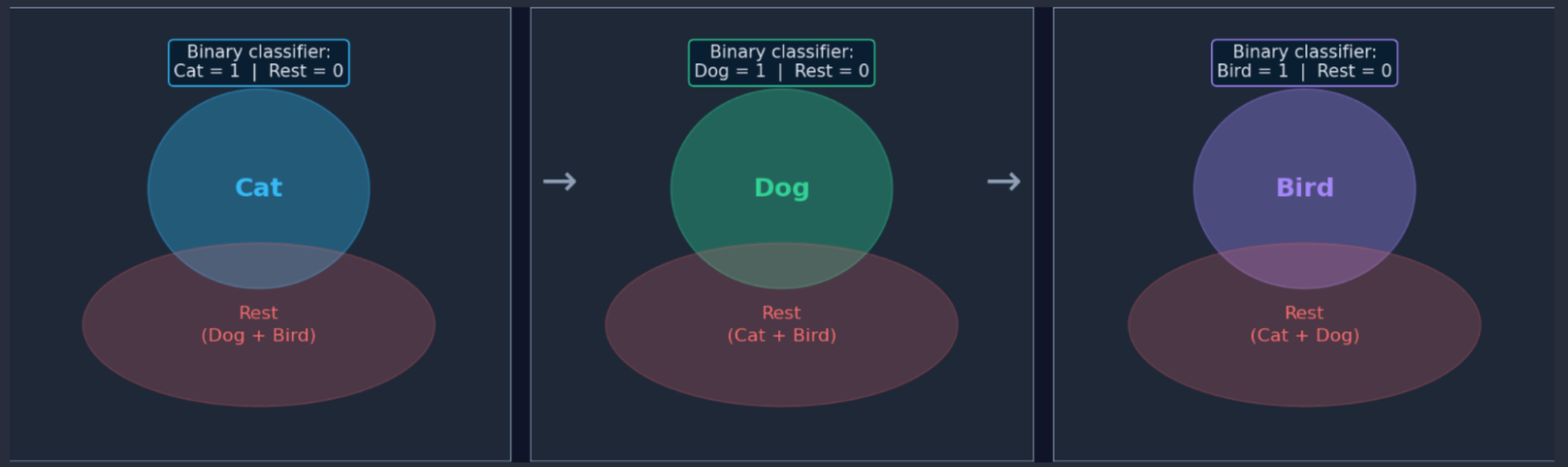
12. Multiclass Evaluation (1/2)

K×K Confusion matrix:
- Diagonal entries: correct predictions per class
- Off-diagonal entries: class confusions (e.g. class i predicted as class j)



12. Multiclass Evaluation (2/2)

Per-class precision, recall, and F1 are computed by treating each class as the positive class under an OvR framing.



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12. Averaging Strategies (1/2)

Example

$$\text{Macro Precision} = (1/K) \cdot \Sigma_k \text{ Precision}_k$$

Each class contributes equally regardless of sample count. Appropriate when performance on minority classes is as important as on majority classes.

$$\text{Weighted Precision} = \Sigma_k \, (\text{support}_k \cdot \text{Precision}_k) \, / \, N$$

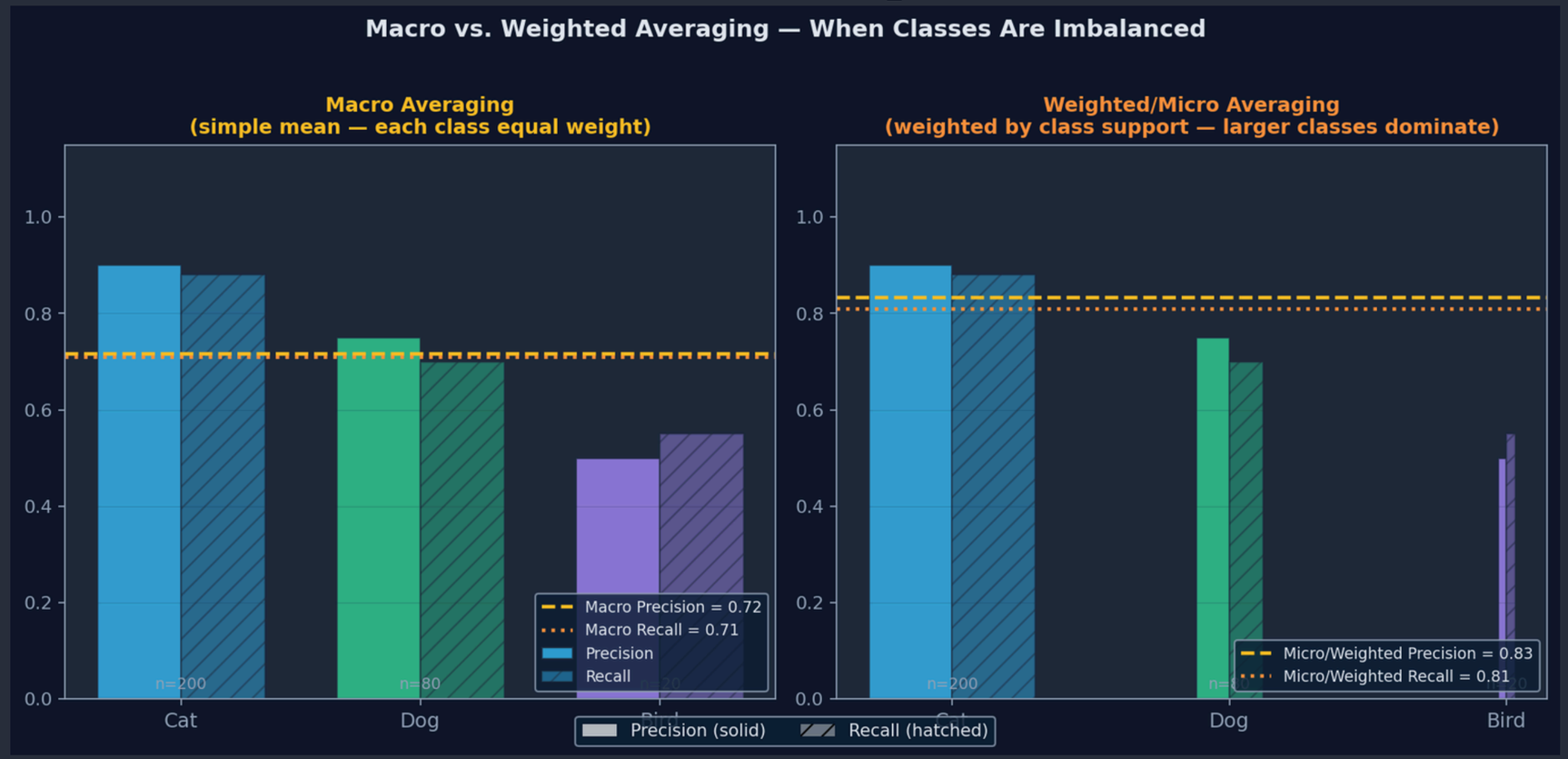
Each class is weighted by its sample count. Reflects overall performance on the observed distribution. Can mask poor performance on minority classes.

$$\text{Micro Precision} = \Sigma_k \, TP_k \, / \, (\Sigma_k \, TP_k + \Sigma_k \, FP_k)$$

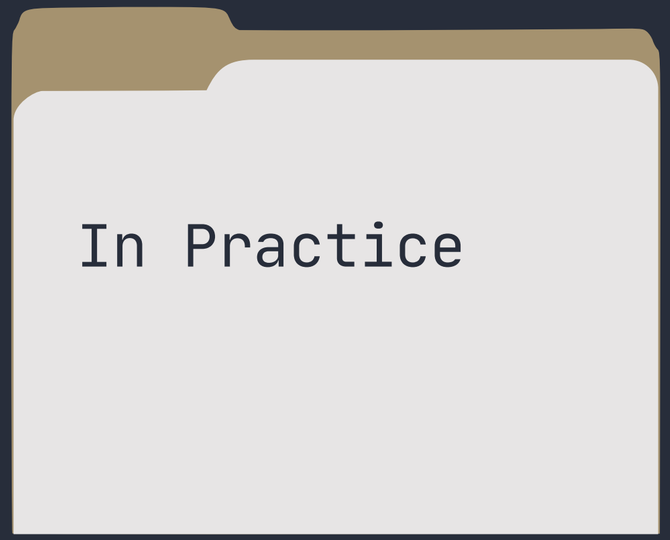
Pools TP/FP/FN counts across all classes before computing the metric.

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12. Averaging Strategies (2/2)



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14. sklearn for evaluation

```
>>> from sklearn.metrics import classification_report
>>> y_true = [0, 1, 2, 2, 2]
>>> y_pred = [0, 0, 2, 2, 1]
>>> target_names = ['class 0', 'class 1', 'class 2']
>>> print(classification_report(y_true, y_pred, target_names=target_names))
              precision    recall  f1-score   support

     class 0       0.50      1.00      0.67         1
     class 1       0.00      0.00      0.00         1
     class 2       1.00      0.67      0.80         3

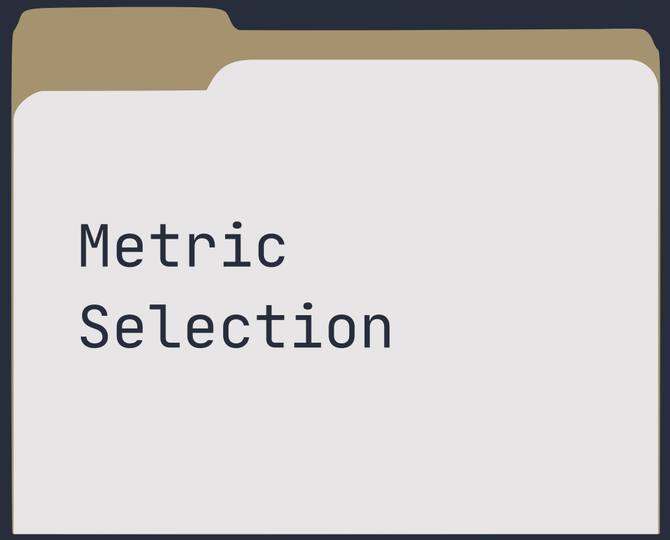
    accuracy                           0.60         5
   macro avg       0.50      0.56      0.49         5
weighted avg       0.70      0.60      0.61         5

>>> y_pred = [1, 1, 0]
>>> y_true = [1, 1, 1]
>>> print(classification_report(y_true, y_pred, labels=[1, 2, 3]))
              precision    recall  f1-score   support

           1       1.00      0.67      0.80         3
           2       0.00      0.00      0.00         0
           3       0.00      0.00      0.00         0

   micro avg       1.00      0.67      0.80         3
   macro avg       0.33      0.22      0.27         3
weighted avg       1.00      0.67      0.80         3
```

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15. Metric Selection

Scenario	Recommended metric	Rationale
Balanced classes	Accuracy or F1	Error costs are symmetric
Imbalanced classes	F1 or AUC	Accuracy is misleading
High FN cost (disease detection, fraud)	Recall, Fβ with β > 1	Minimise missed detections
High FP cost (spam filter, alerts)	Precision, Fβ with β < 1	Minimise false alarms
Multiclass, imbalanced	Macro F1	Prevents majority classes from dominating the score

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Any Questions ?

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Thank you for your attention

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